ScienceDirect

Journal of Power Sources

Volume 660, 30 December 2025, 238524

Effective water management through microporous-layer wettability control for high power density in low-temperature fuel cells

DoEun Kim [a], Taeyang Han [c], JunYoung Seo [a], HangJin Jo [a b]

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https://doi.org/10.1016/j.jpowsour.2025.238524

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Highlights

•MPL wettability was modified using plasma beam and NaOH solution.

•Modifying only the top surface of MPL was sufficient to improve water management.

- Differences in fuel-cell performance were observed depending on hydrophilic ratio.

- Water management in MPL affected charge-transfer and mass-transfer resistance.

Abstract

Water management in microporous layers (MPLs) is a major challenge in low-temperature fuel cells. This study introduces a simple method to fabricate a biphilic MPL with in-plane hydrophilic and hydrophobic regions. The hydrophilic ratio is systematically controlled to analyze its effect on cell performance. Surface wettability is selectively modified via plasma treatment and subsequent NaOH treatment, while preserving the original structure. Contact-angle measurements and energy-dispersive X-ray spectroscopy confirm successful modification. Among the tested configurations, a hydrophilic ratio of 25.0% exhibits the most favorable performance among the tested conditions: a 28.8% increase in current density at 0.6V and 22.2% increase in maximum power density relative to the hydrophobic MPL. Electrochemical impedance spectroscopy reveals improved mass transfer in the biphilic structure. A 5-h water-drainage test confirms its superior water-removal capability. Comparative analysis with previous studies employing similar surface modification techniques further demonstrates enhanced performance of the proposed MPL, which exhibits the largest improvement in current density at 0.6V among reported results. These findings indicate that in-plane wettability patterning promotes liquid–gas pathway separation. The proposed method is simple, scalable, and applicable for improving liquid–gas management and performance in low-temperature fuel cells.

Previous

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Keywords

water management; biphilic; Microporous layer; Low-temperature fuel cell; Flooding

1. Introduction

Low-temperature fuel cells have emerged as indispensable technologies across various industries because of their rapid response to fluctuations in renewable energy output and their high energy density [1,2]. They are widely utilized in transportation [3], portable electronics [4], stationary power systems [5], and aerospace applications [6] and play a crucial role as ecofriendly and sustainable energy sources. Widely used low-temperature fuel cells include polymer electrolyte membrane fuel cells (PEMFCs) [7,8], alkaline fuel cells (AFCs) [9,10], direct methanol fuel cells (DMFCs) [11,12], and anion exchange membrane fuel cells (AEMFCs) [13,14], each optimized for different use cases. These technologies are key enablers of current and future energy transitions toward sustainability.

At temperatures below 100°C, low-temperature fuel cells exhibit unique water behavior [15]. At these temperatures, the water produced as a byproduct of the electrochemical reactions remains in the liquid phase and is critical for sustaining membrane hydration, ensuring high proton conductivity. Although proper membrane hydration is essential for efficient fuel-cell operation, excessive water accumulation in the catalyst layer (CL) and gas diffusion layer (GDL) can impede reactant gas transport and degrade the overall performance. The GDL, which is located between the CL and flow channel, comprises a microporous layer (MPL) and a gas diffusion backing layer (GDBL), with distinct pore-size distributions at the nanoscale and microscale, respectively.

Accordingly, effective water management in the GDL involves not only controlling the amount of water but also achieving efficient liquid–gas path separation to simultaneously promote membrane hydration and reactant gas transport. Therefore, the design of the GDL plays a critical role in maintaining distinct pathways for liquid-water removal and reactant gas transport, both of which are essential for optimizing fuel-cell performance [[16], [17], [18], [19], [20]]. Various approaches have been explored to enhance the water management in GDLs by modifying their physical structure and wettability. Among the structural modification strategies, the porosity gradient method has been investigated for its effect on liquid water transport and distribution using X-ray visualization [21], the lattice Boltzmann method [22], and the volume of fluid (VOF) method [23]. In addition, the interaction between heat and water transport has also been examined through numerical simulations [24]. The laser perforation technique has been reported to facilitate liquid water discharge and improve cell performance by introducing preferential drainage pathways [[25], [26], [27], [28], [29]]. X-ray radiography and tomography further confirmed its influence on water distribution within the GDL [30,32,33], while numerical studies highlighted reduced two-phase transport resistance and improved oxygen accessibility [31,34]. However, the porosity gradient approach is limited by the complexity of its fabrication processes, and local structural damage around perforations has also been observed in the laser perforation technique, which may adversely affect water management [25,32]. As an alternative, the patterned or dual-wettability approach, which selectively modifies surface wettability without altering the physical structure, has attracted increasing attention. By separating gas and liquid transport pathways, this method provides a promising route for improving water management while preserving the structural integrity of the GDL, and its effectiveness has been demonstrated through synthetic routes and wettability tuning [35,36], neutron and X-ray imaging [[37], [38], [39]], as well as performance evaluations under patterned wettability designs [[40], [41], [42], [43]]. However, applying dual-wettability treatments across the entire thickness of the GDL, from the MPL to the flow channel, can increase the droplet detachment diameters in hydrophilic regions, leading to flooding at the interface between the flow channel and GDBL [44]. Moreover, owing to the small pore size of the MPL, the highest breakthrough pressure is required on the MPL side. Therefore, instead of modifying the entire GDL, recent studies have focused on partially engineering the MPL with dual wettability, which can improve water management by facilitating the separation of the gas and liquid transport pathways.

Several methods have been explored for the selective hydrophilic modification of the MPL. Doping various hydrophilic materials into the MPL, such as hydrophilic phenolic resin (PR) and microcrystalline cellulose (MCC) [45], hydrophilic carbon nanotubes [46], and aluminosilicate fibers [[47], [48], [49]], have proven effective in enhancing water management and increasing cell voltage, even under high-current density conditions. Spernjak et al. demonstrated that MPLs containing multi-walled carbon nanotubes (MWCNTs) and aluminosilicate fibers had superior water-removal capability from the CL, reducing the mass-transfer resistance by 50%–80% at 1.4 A/cm^2 compared with conventional carbon black-based MPLs [48]. In addition to the choice of the material, the amount of dopant is a key factor affecting water management. Kitahara et al. identified the optimal amount of hydrophilic dopant by systematically adjusting its concentration. Under both low- and high-humidity conditions, the addition of 4 mass% hydrophilic CNTs resulted in the best cell performance [46]. However, the random distribution of the hydrophilic and hydrophobic domains often results in inconsistent performance across fuel cells, hindering their scalability and commercial viability. This highlights the need for a more controlled and reliable strategy for engineering the MPL to enhance water transport, while maintaining cell performance and manufacturability. Although there have been reports demonstrating that well-controlled biphilic MPLs can enhance fuel-cell performance, discussions on the biphilic parameters, specifically the optimal ratio of hydrophilic areas remain limited. For example, Wang et al. developed a dual-wettability MPL with a hydrophilic content of approximately 50% and evaluated its performance under both cold-start and normal operating conditions [50,51]. Although the dual-wettability MPL proved effective in improving the cold-start performance, it did not exhibit significant performance gains under normal conditions. Jia et al. applied a well-aligned biphilic MPL with wettability patterning to an MEA employing an ultra-thin catalyst; however, a systematic study on the optimal fraction of hydrophilic regions was not conducted [52].

In this study, we developed a simple and scalable method for fabricating dual-wettability (biphilic) MPLs and systematically investigated the effect of in-plane wettability patterning by varying the hydrophilic ratio (12.5%, 25.0%, 37.5%, 50.0%, and 75.0%) as part of ongoing efforts to enhance the accessibility of fabrication methods and systematic parametric studies on biphilic MPL designs. A commercial hydrophobic GDL with an MPL was used as a 0% reference. Electrochemical polarization curves and electrochemical impedance spectroscopy (EIS) were employed to evaluate the fuel-cell performance and the effects of the hydrophilic ratio on the mass and charge transfer characteristics. The effectiveness of surface treatment was confirmed through complementary experiments.

2. Experimental

2.1. Fabrication of biphilic MPLs

Biphilic MPLs were fabricated by modifying a commercially available GDL (JNT20-A6H) with a coated MPL. According to the results of Wang et al., when the hydrophilic ratio was 50% in the MPL, there was little difference in performance compared with a fully hydrophobic MPL under normal operating conditions [49,50]. Accordingly, in this study, the hydrophilic area ratio was varied (12.5%, 25.0%, 37.5%, 50.0% and 75.0%) to investigate its impact on fuel cell performance, the results of which are shown in Table 1. To systematically investigate the performance trends, the hydrophilic ratio was varied by ±12.5%–25.0%. The fabrication process for the biphilic MPL is illustrated in Fig. 1(a). To control the hydrophilic ratio, an Al mask with five regularly spaced openings was fabricated by laser-cutting a thin aluminum sheet. These openings defined the areas exposed to plasma irradiation, determining the resulting hydrophilic regions. The mask was placed on the top surface of the MPL, and atmospheric-pressure plasma (80W, 10 sccm) was applied for 10min. Subsequently, the plasma-treated MPL was briefly immersed in a 50wt% NaOH solution (Sigma-Aldrich, USA) and removed immediately. During this process, the NaOH solution was bonded to the surface of the MPL modified by the plasma treatment. The sample was kept at room temperature under ambient conditions for approximately 24h with residual NaOH on the surface, which led to the peeling of the hydrophobic coating (polytetrafluoroethylene, PTFE). After these processes, the samples were rinsed with deionized water to remove any residual NaOH. The biphilic structure formed a stripe pattern, and the stripes were oriented perpendicular to the flow channel and rib directions. Following these processes, a biphilic MPL was formed, as shown in Fig. 1(b) and (c).

Table 1. Design parameters of the laser-cut aluminum mask for controlling the hydrophilic ratio.

Hydrophilic ratio	**p [mm]**	**w [mm]**
12.5%	4	0.5
25.0%	4	1
37.5%	4	1.5
50.0%	4	2
75.0%	4	3

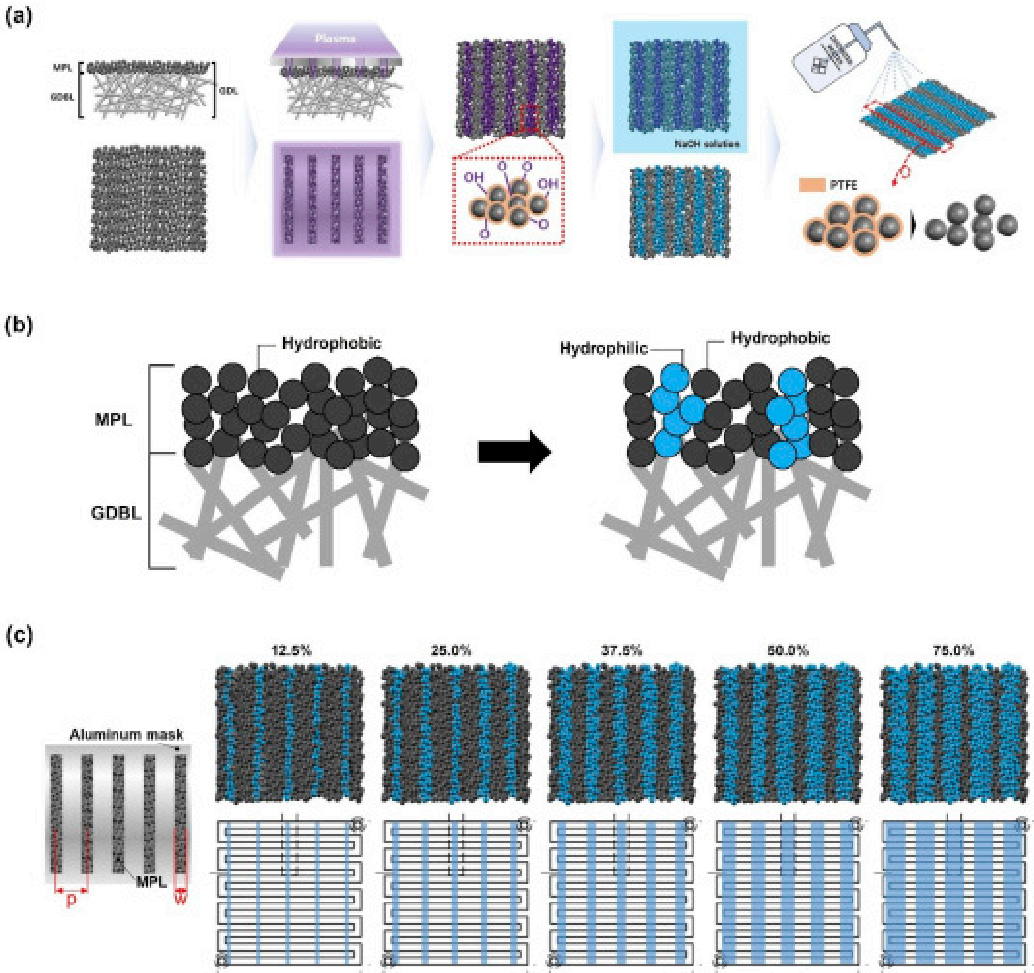

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Fig. 1. (a) Schematic of the fabrication process for the modified biphilic MPL. (b) Schematic illustrating the modification of a commercial hydrophobic MPL (left) into a biphilic structure (right) through the introduction of patterned hydrophilic regions (blue). (c) Biphilic MPLs designed with varying hydrophilic ratios. The biphilic stripes were oriented perpendicular to the channel and rib directions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.2. Surface characterization of MPLs

To evaluate the effect of surface modification on the wettability, only half of each MPL sample was subjected to surface treatment, whereas the other half remained unmodified. The surface water contact angles were measured using the sessile-drop method on both the MPL and GDBL sides of the modified and unmodified regions, as shown in Fig. 2(a). A 5-μL water droplet was dispensed onto each region, and water contact angles were measured on both the left and right sides of the droplet. The measurements were performed for at least five droplets per region, and the results were averaged.

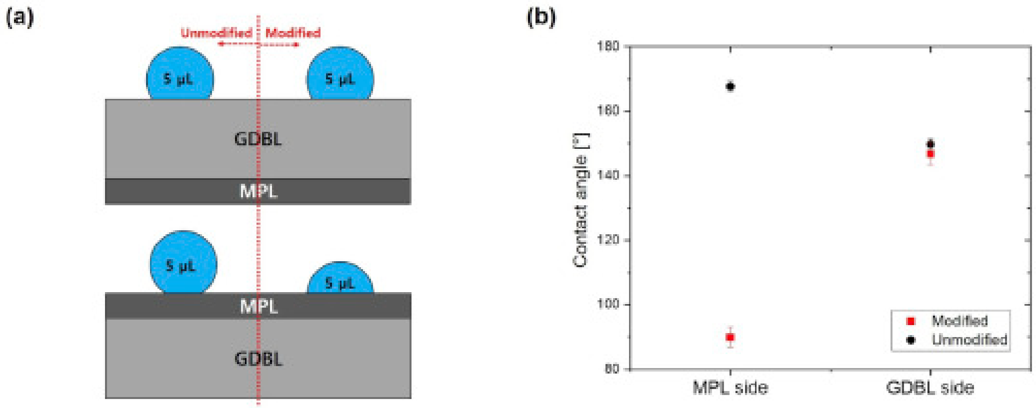

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Fig. 2. Contact angles of GDLs: (a) schematic of the contact-angle measurement. (b) Measured contact angles on the MPL and GDBL sides.

The contact-angle measurements are shown in Fig. 2(b). The contact angle of the unmodified MPL was 167.6°, whereas that of the modified MPL was 89.9°. This indicates that the surface treatment effectively removed the PTFE coating, exposing the underlying C particles, whose intrinsic wettability, which is characterized by relatively low contact angles (~90°), has been previously reported [53], confirming the effectiveness of the treatment. In contrast, the contact angle of the GDBL, which was on the opposite side of the modified part, was almost identical, regardless of the MPL surface treatment. This suggests that the plasma treatment effectively changed only one side while preserving the other, achieving selective surface modification.

In addition, energy-dispersive X-ray spectroscopy (EDS) analysis was conducted on the surface of the MPL at an accelerating voltage of 5 kV to verify the coexistence of the hydrophobic and hydrophilic regions, as shown in Table 2 and Fig. S3. The fabrication process of the biphilic MPL for EDS analysis and the analysis regions are provided in Fig. S1–S2. Half of the MPL was left unmodified, whereas the other half was subjected to surface treatment. Because the chain structure of PTFE consists of C and F [54], the modified MPL regions were expected to exhibit a lower F content, corresponding to the areas where the PTFE coating had been removed. Consistent with this expectation, Table 2 indicates that the mass percentages of C and F differed between the unmodified and modified regions. There were differences between the modified and unmodified MPLs; however, they were small because EDS collects composition data from not only the surface but also a certain depth beneath the surface. This is attributed to the fact that only the top surface of the MPL adjacent to the CL was selectively treated, whereas the EDS technique reflected the average composition over a relatively large interaction volume. This is attributed to the fact that only the top surface of the MPL adjacent to the CL was selectively treated, whereas the EDS technique reflected the average composition over a relatively large interaction volume. Therefore, as shown in Fig. S1, no pronounced difference was observed in the EDS images. However, since the analysis depth of EDS is generally known to be 0.02–1.0 μm [55], the treatment depth is conservatively estimated to be less than 0.02 μm, indicating that only an extremely shallow layer was modified. This interpretation is supported by the observations in Fig. 3. A portion of the already modified biphilic MPL sample was peeled off using adhesive tape, which physically removed the modified surface particles from the region. After the sample was immersed in water and removed, the peeled-off region lost its biphilic characteristics, exposing the underlying hydrophobic particles. In contrast, other regions retained their biphilic characteristics.

Table 2. Contents (mass%) of C and F in modified and unmodified regions.

		C (mass%)	F (mass%)
Low-magnification (100×)	unmodified	83.09 ± 0.12	16.91 ± 0.13
	modified	87.70 ± 0.12	12.30 ± 0.11
High-magnification (10000×)	unmodified	93.64 ± 0.10	6.36 ± 0.07
	modified	96.06 ± 0.09	3.94 ± 0.05

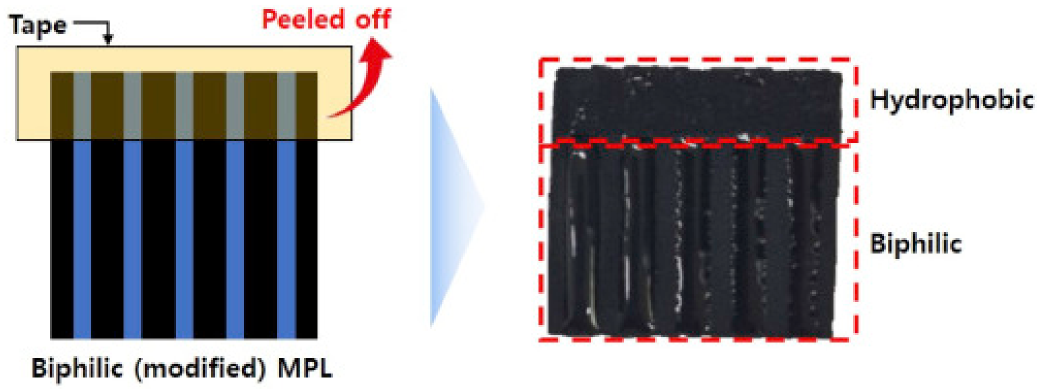

Fig. 3. Water-immersion test of the biphilic MPL after tape peeling.

2.3. Single-cell tests

As shown in Fig. 4, the Nafion 211 membrane was coated with Pt as a catalyst, and its contents were 0.4 mg/cm^2 for both the anode and cathode, with an active area of 4 cm^2 (2 cm×2 cm). A commercial GDL (JNT20-A6H, with MPL) was used in its unmodified, hydrophobic state at the anode. The same GDL was surface-modified to a biphilic MPL and applied at the cathode, as flooding is a critical issue primarily on the cathode side, where water is generated during fuel-cell operation. The flow rates of the reactants were fixed at 100 and 300 sccm for the anode and cathode, respectively. The flow channel was a single-serpentine design, with a channel width of 1.0 mm, rib width of 0.9 mm, and channel depth of 0.32 mm. These flow rates were intentionally set in excess of the stoichiometric requirements to ensure a uniform distribution of reactants across the entire electrode without depletion and to minimize the influence of reactant starvation. The relative humidity (RH) of the supplied gases was set to 100% for both the anode and the cathode to fully hydrate the membrane and create a high-humidity environment that easily induced flooding. This condition was selected to evaluate the water-management capability of the biphilic MPL under flooding scenarios. The RH was adjusted by changing the temperature of the humidifier, while the cell temperature was maintained at 60 °C, which is the typical operating condition for PEMFCs. The performance of the fuel cell was measured in the constant-current mode, with the current density increased in steps of 0.1 A/cm^2. After the polarization curves were measured, corresponding EIS measurements were performed to evaluate the electrochemical performance of the fuel cell. EIS was performed at 0.6V with a 5-mV amplitude over the frequency range of 100kHz to 0.1Hz.

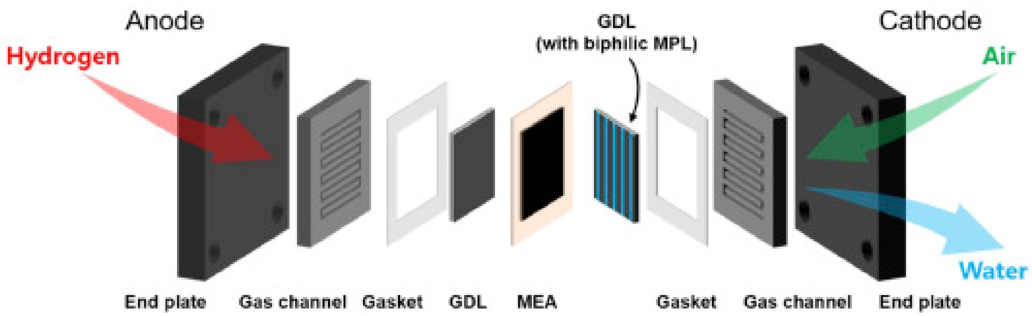

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Fig. 4. Schematic of the single cell.

3. Results and discussion

3.1. Performance of single cell with modified MPLs

The polarization curves and performance enhancements of the MPLs with different hydrophilic ratios are shown in Fig. 5. Notably, although only the top surface of the MPL adjacent to the CL was modified to form biphilic regions, significant changes in cell performance were observed with the ratio of surface wettability. As shown in Fig. 5, the performance of the cell depended significantly on the hydrophilic ratio. As indicated by Fig. 5(b), in the case of a low hydrophilic ratio (12.5%), the performance exhibited negligible differences compared with that of the hydrophobic MPL, indicating that the surface modification was insufficient to adequately change the water-transport behavior. In contrast, the samples with hydrophilic ratios of 25.0% and 37.5% demonstrated performance enhancements, as shown in Fig. 5(c), indicating that a proper balance between water removal and gas transport was achieved. Specifically, the 25.0% hydrophilic ratio exhibited the best performance among the samples, with a 28.8% increase in current density at 0.6V and a 22.2% increase in maximum power density, while the 37.5% hydrophilic ratio also exhibited appreciable improvement, with increases of 17.2% and 8.40%, respectively. However, higher hydrophilic ratios led to performance degradation. As shown in Fig. 5(d), the 50.0% hydrophilic ratio sample exhibited slight performance degradation, and the 75.0% hydrophilic ratio sample exhibited the lowest performance among all the tested samples, with a 12.9% reduction in current density at 0.6V and a 12.5% reduction in maximum power density compared with the hydrophobic MPL. This decline is attributed to excessive water accumulation caused by the hydrophilicity of a large portion of the MPL, which likely caused flooding and hindered gas diffusion.

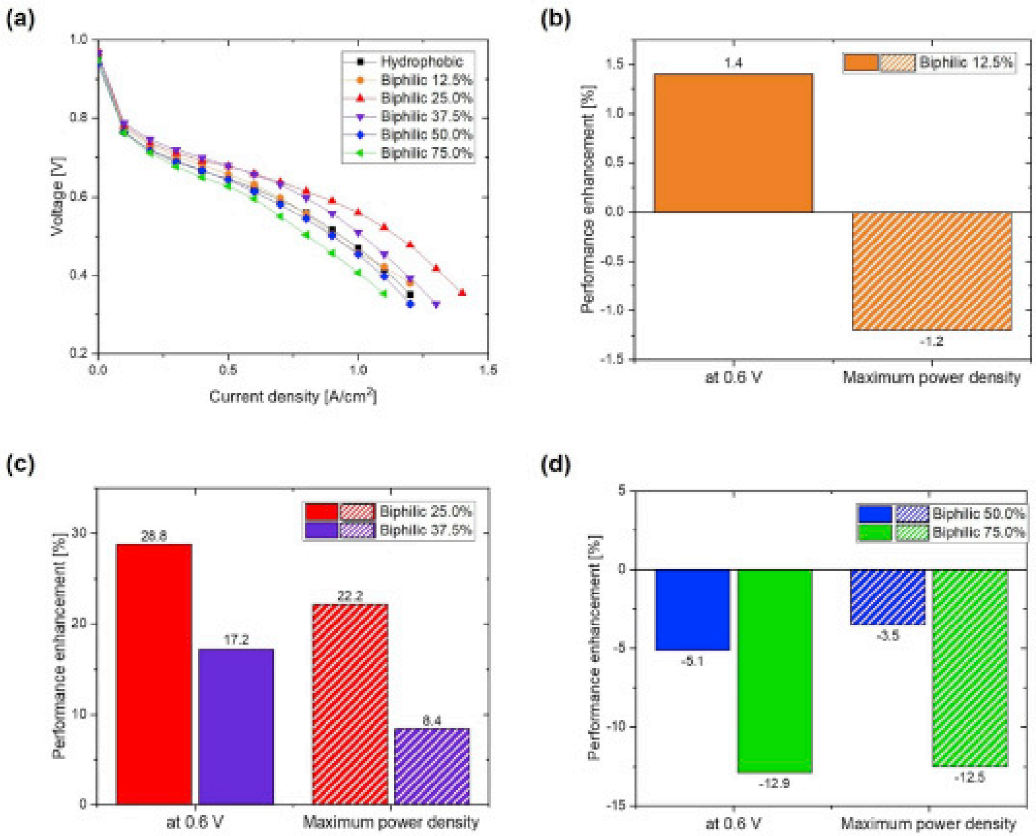

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Fig. 5. Performance of MPLs for each hydrophilic ratio. (a) Polarization curves of MPLs. (b) Performance enhancements of the 12.5% hydrophilic ratio MPL, (c) 25.0% and 37.5% hydrophilic ratio MPLs, and (d) 50.0% and 75.0% hydrophilic ratio MPLs compared with the hydrophobic MPL.

To investigate the performance differences, EIS analysis was conducted. Fig. 6(a) and (b) present the EIS results at 0.6V with respect to the hydrophilic ratio. The data were fitted using a simple equivalent circuit, i.e., R_{HFR}(R1Q1)(R2Q2), where R_{HFR} represents the membrane and contact resistances (R_{Ω}), R1 primarily corresponds to the kinetic resistance (R_{ct}), R2 corresponds to the mass-transfer resistance (R_{mt}), and Q1 and Q2 denote constant-phase elements to enable the modeling of non-ideal capacitive behavior in porous electrodes and surface roughness. The fitting circuit is also illustrated in Fig. 6(a). According to the EIS analysis, no significant differences in ohmic resistance (R_{Ω}) were observed across all hydrophilic ratios, including the hydrophobic MPL. This was likely due to the consistent assembly pressure applied to all cells, which minimized the variation in contact resistance, and the consistent feeding of fully humidified reactants, which ensured uniform membrane hydration across all tests.

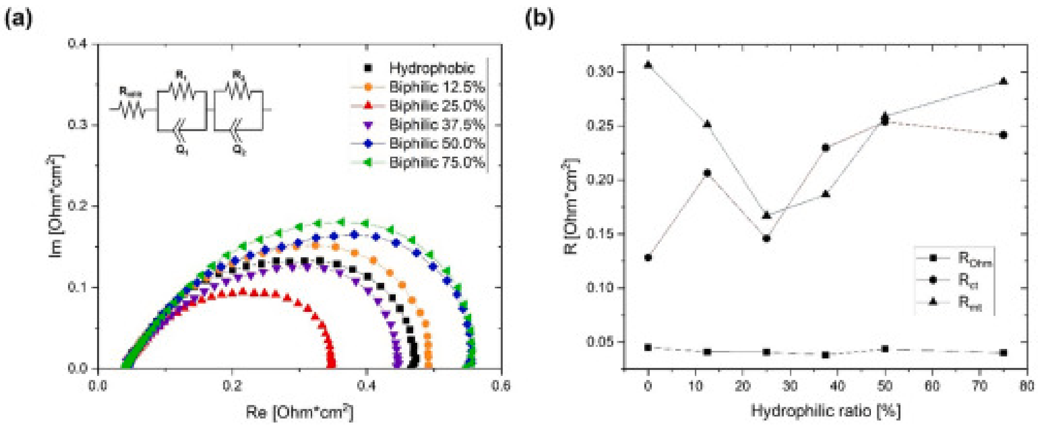

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Fig. 6. (a) EIS results at 0.6V with fitted circuits. (b) Contribution of each equivalent circuit component for different MPLs; variation tendencies of the ohmic resistance (R_{ohm}), charge-transfer resistance (R_{ct}), and mass-transfer resistance (R_{mt}).

The charge-transfer resistance (R_{ct}) depended on the hydrophilic ratio, with values increasing by up to 13.8% (at a hydrophilic ratio of 25.0%) and reaching a maximum increase of 98% (at a hydrophilic ratio of 50.0%) relative to the hydrophobic MPL. As reported by Avcioglu et al., R_{ct} is affected by water management in the CL [56]. Moreover, a previous study reported an expression for activation loss induced by liquid water saturation(s) and this reduces the effective reaction area and leads to an increase in

Rct as below Equation (1) [57].

$$\Delta\eta_{act}^{lw} = \frac{RT}{\alpha nF}\ln\left(\frac{1}{1-s}\right) \tag{1}$$

The observed increase in R_{ct} with the hydrophilic MPL treatment was likely due to the accumulation of liquid water beneath the modified regions, which may have resulted in localized catalyst deactivation. For the 12.5% hydrophilic ratio, because of the insufficiently defined hydrophilic region, water may have been randomly distributed outside the intended area, contributing to the increase in R_{ct}. When the hydrophilic ratio was 25.0%, the R_{ct} value increased only slightly, indicating that the catalyst kinetics under these conditions were comparable to those of the hydrophobic MPL. Moreover, beyond a hydrophilic ratio of 25.0%, the R_{ct} value began to increase at 37.5%. When the hydrophilic ratios were 50.0% and 75.0%, it was presumed that the excessively defined hydrophilic regions caused the CL to be covered by a significant amount of liquid water, deactivating a large portion of the catalyst sites. The absence of a significant difference in the R_{ct} values between the 50.0% and 75.0% cases suggests that beyond a certain threshold (50.0%), the amount or distribution of water in the CL remains largely unchanged, resulting in similar electrochemical behavior. This suggests that an appropriate level of hydrophilic modification minimizes the extent of deactivation within the CL.

The mass-transfer resistance (R_{mt}) primarily increased owing to insufficient simultaneous water and gas management in the GDL. For the sample with a hydrophilic ratio of 12.5%, the R_{mt} was reduced by 18% compared with that of the hydrophobic MPL. This indicates a partial improvement in water management; however, it was not the greatest reduction in R_{mt} among all the tested samples, suggesting that effective separation of gas and liquid pathways did not occur. This result implies that although partial water removal was achieved through the hydrophilic regions, liquid water may also spread beyond these areas, leading to a non-uniform water distribution. This increases the tortuosity of the gas pathways, which likely hinders gas transport. In contrast, the sample with a 25.0% hydrophilic ratio exhibited the greatest reduction in R_{mt}, which was 45.5% lower than that of the hydrophobic MPL, along with the most effective separation of liquid and gas pathways, among the tested samples. This result suggests that the appropriate hydrophilic ratio not only forms effective water-removal pathways but also maintains stable and sufficient gas-transport pathways. The sample with a hydrophilic ratio of 37.5% also exhibited a significant reduction in R_{mt} (39%), indicating that hydrophilic patterning remained effective in supporting water removal. However, its performance was inferior to that of the 25.0% hydrophilic ratio sample, which exhibited the greatest R_{mt} reduction. This suggests that although a hydrophilic ratio of 37.5% still facilitated water removal, it may have begun to interfere with the formation of sufficient gas pathways. Meanwhile, for the 50.0% hydrophilic ratio sample, R_{mt} was reduced by 15.5% compared with that of the hydrophobic

MPL. This relatively small improvement is attributed to the excessive water retention and insufficient formation of gas pathways due to the large hydrophilic area. These findings highlight the importance of maintaining effective gas path management, in addition to managing liquid-water pathways. For the sample with a hydrophilic area ratio of 75.0%, the R_{mt} value was comparable to that of the hydrophobic MPL, suggesting that the water distribution within the GDL behaved similarly to that in the hydrophobic case. This indicates that at such a high hydrophilic ratio, the GDL exhibits almost uniform wettability.

3.2. Drainage capability of biphilic MPL

To further demonstrate that the modified MPL has an enhanced water-management capability, it was tested under 5-h constant-voltage discharges at 0.6V. Among biphilic MPLs with various hydrophilic ratios, the 25.0% hydrophilic ratio, which exhibited the best performance, was selected for testing. Previous studies have indicated that the accumulation of liquid water can block gas pathways, leading to a sudden increase in local pressure. Once the excess water is expelled, the output voltage decreases sharply, followed by a rapid recovery. As shown in Fig. 7, although run at a higher current density, the modified MPL with a hydrophilic ratio of 25.0% exhibited better water management. Over a 5-h period, four instances of water release were observed for the biphilic MPL, whereas only two were recorded for the hydrophobic MPL. This suggests that in the hydrophobic MPL, liquid water accumulates randomly, resulting in a longer and more tortuous path for the product water to be discharged. In contrast, in a well-modified MPL, the presence of well-defined hydrophilic pathways shortens the liquid transport path, leading to more frequent water discharge.

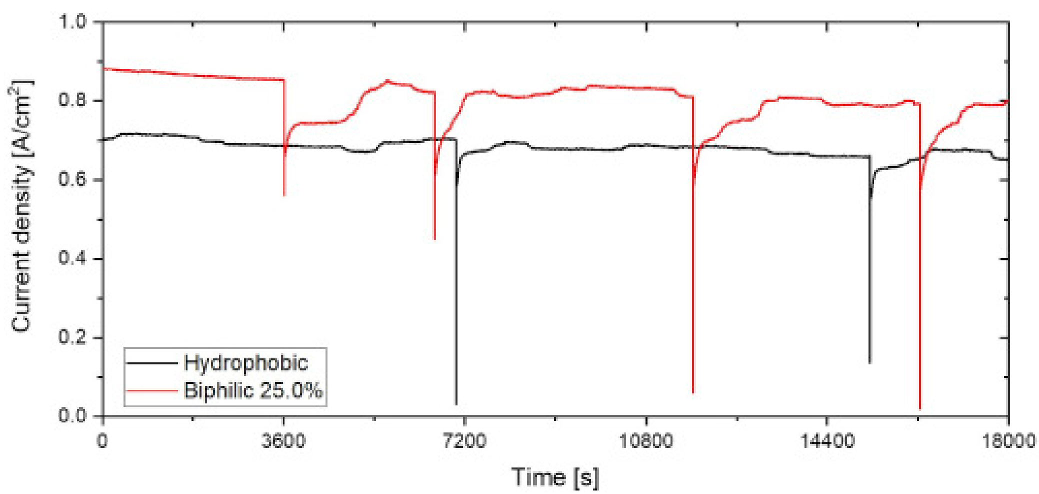

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Fig. 7. Antiflooding tests at 0.6V with the biphilic 25.0% MPL and commercial hydrophobic MPL.

3.3. Comparison with previous studies

We compared the biphilic MPLs prepared in this study with reported previously in terms of their current densities at 0.6V and maximum power density (Fig. 8). The details of the samples used for comparing performance improvement are presented in Table 3. Among these, the greatest current-density enhancement at 0.6V was observed in the present study when the hydrophilic area ratio was 25.0%. While previous studies typically assessed the performance at fixed or limited hydrophilic ratios, this study conducted a systematic parametric investigation by varying the hydrophilic ratio across a broad range to investigate its quantitative impact on fuel-cell performance. This result suggests that precise control of the hydrophilic ratio significantly improves the water-management capability and that biphilic pattern optimization can effectively maximize cell performance.

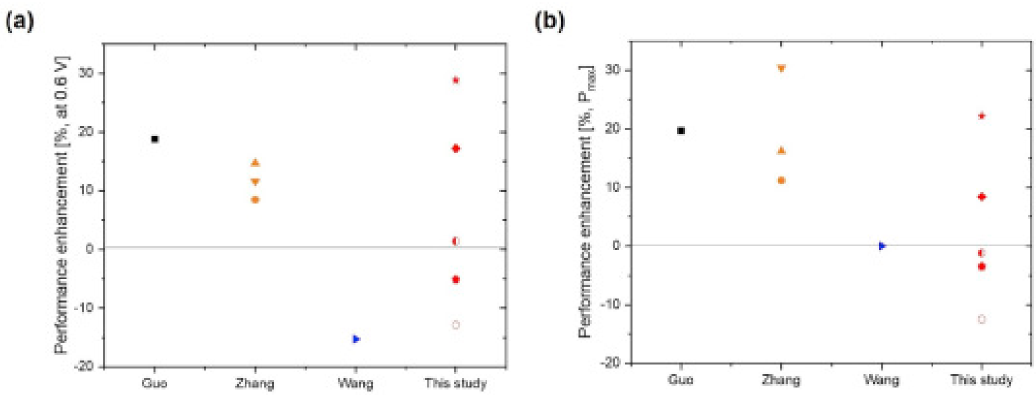

Fig. 8. Comparative analysis of performance enhancements achieved by wettability-engineered MPLs in this study and previous studies. (a) Performance enhancement at 0.6V and (b) enhancement in maximum power density (P_{max}).

Table 3. Details regarding the samples used in the comparative analysis of performance enhancement (Fig. 8), including hydrophilic ratios, modified regions, and corresponding references.

	Symbol	Hydrophilic ratio	Modified region	Ref.
Guo	■	MPL+GDBL	MPL+GDBL	[42]
Zhang	Pattern A: ●	Pattern A: ~2.3%	MPL+GDBL	[43]
	Pattern B: ▼	Pattern B: ~4%		
	Pattern C: ▲	Pattern C: ~9%		
Wang	▶	50%	MPL	[50]
This study	12.5%: ◐	12.5%, 25.0%, 37.5%, 50.0%, 75.0%	MPL	
	25.0%: ★			
	37.5%: ◆			
	50.0%: ⬟			
	75.0%: ○			

Interestingly, Zhang et al.'s Pattern B, which featured a hydrophilic ratio of approximately 4%, achieved the greatest enhancement in the maximum power density, with an improvement of approximately 30% [43]. The second-best enhancement was observed in our study at a hydrophilic ratio of 25.0%. This difference is attributed to variations in the structural properties of the MPL and GDBL, such as porosity and thickness, which can affect the performance in the mid- and high-current-density regions.

Another key observation was that the optimal hydrophilic ratio for performance enhancement in our study was significantly higher than that reported by Zhang et al. [43]. This discrepancy likely arises from the difference in the modified layers; while Zhang et al. applied biphilic treatment to both the MPL and GDBL, our study limited the modification to the MPL. When the GDBL is rendered hydrophilic, a larger droplet detachment diameter may result, hindering reactant transport. Consequently, in such cases, the optimal performance tends to occur at low hydrophilic ratios. Conversely, because our approach targeted only the MPL, a higher hydrophilic area ratio was needed to achieve the optimal performance compared with the case where both the MPL and GDBL were rendered biphilic.

It should also be recognized that the observed differences among studies are not solely attributable to the properties of the MPL or GDBL themselves. Device-level parameters, including the overall cell size and flow-field configuration, can also influence how effectively a given water-management strategy performs. In addition, the effect of cell size may also play a role in determining the effectiveness of water-management strategies. Marappan et al. [58] reported that enlarging the reaction area from 25 to $100 cm^2$ reduced the relative improvement achieved by porous sponge inserts, largely due to the increased pressure drop associated with scale-up. Similarly, Xu et al. [59] showed that a flow field with capillary microchannels improved performance in both 25 and $100 cm^2$ cells, but the degree of enhancement decreased slightly in the larger cell, emphasizing the role of flow-field design in compensating for pressure losses. These findings indicate that performance gains achieved through water-management strategies are not only a function of the intrinsic properties of the GDL/MPL but also depend on device-level parameters such as cell size and channel configuration. Although the present work was conducted using a small-area ($4 cm^2$) single cell to enable systematic comparison of hydrophilic area ratios under well-controlled conditions, whether the same relative benefits can be realized in larger-area cells remains an important question. Nevertheless, the results of this study remain significant, as they clearly demonstrate the effect of the hydrophilic area ratio under rigorously controlled conditions. This underscores the need for future studies to validate the scalability and practical applicability of the proposed biphilic MPL design.

4. Conclusions

This study proposed a novel method for fabricating biphilic MPLs using a commercial GDL; to the best of our knowledge, this method has not been previously reported. The top surface of the MPL was successfully modified, and significant differences in cell performance were observed. We aimed not to determine the optimal hydrophilic ratio but rather to systematically investigate how varying the hydrophilic ratio affects fuel-cell performance through a parametric study. The polarization curves indicated that a hydrophilic ratio of 25.0% yielded the highest performance among the tested ratios, significantly enhancing mass transport while having minimal impact on reaction kinetics. These results indicate that in addition to managing liquid-water pathways, ensuring sufficient gas-transport pathways is essential for enhancing fuel-cell performance. Furthermore, the hydrophilic MPL, which exhibited the highest performance among the tested hydrophilic ratios, promoted the formation of shorter liquid water pathways. This well-defined separation of the liquid and gas transport routes facilitates the efficient removal of accumulated water within the cell. Compared with other studies employing dual-wettability strategies, the proposed biphilic MPL fabrication and hydrophilic ratio control method achieved superior performance enhancement at 0.6V. The improvement in the maximum power density was the second-best among the referenced studies. This may be partly attributed to structural characteristics such as the porosity and thickness of the MPL and GDBL. However, we acknowledge that other factors, including the overall cell size and the design of the flow channels, could also influence the observed performance differences. These uncertainties suggest that the effect cannot be solely ascribed to the structural parameters, and further systematic studies are required to decouple these contributions. These parameters likely determine whether the optimal performance is achieved in the medium- or high-current-density operating regimes. Moreover, the results suggest that when only the MPL is functionalized with a biphilic structure, as in this study, a higher hydrophilic area ratio is needed to attain optimal performance compared with configurations in which both the MPL and GDBL are modified.

In future work, we aim to gain a more inclusive understanding of the pore-size and wettability distributions within the MPL. To this end, we plan to conduct further analyses of key structural parameters, including pore size, porosity, MPL thickness, and the depth of hydrophilic surface treatment, and to quantitatively relate these parameters not only to mass-transport resistance and flooding behavior but also to charge transfer resistance, since the biphilic structure is expected to influence both transport and kinetics. In addition, the water-management performance of MPL should be evaluated under various operating conditions to better reflect real-world applications. At the same time, we will systematically decouple device-level effects by varying cell size and flow-field configuration (e.g., channel/rib width and depth, serpentine layout) while holding MPL properties constant. Conversely, we will also vary MPL properties while keeping the flow-field configuration fixed. Through this approach, we aim to isolate the contributions of intrinsic MPL/GDBL structure from geometry-driven effects. Furthermore, current PEMFC industry targets are typically 5000 h for automobile, 20,000 h for bus, and up to 40,000 h for stationary power [60]. Therefore, it is essential to include long-term endurance tests of both treated and untreated MPLs under wet conditions, as the treated MPLs are expected to improve durability through enhanced water-management capability, which we aim to investigate.

CRediT authorship contribution statement

DoEun Kim: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Taeyang Han:** Writing – review & editing, Methodology, Conceptualization. **JunYoung Seo:** Writing – review & editing, Investigation. **HangJin Jo:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding: This work was supported by the Institute for Korea Spent Nuclear Fuel (iKSNF) and a Korea Foundation of Nuclear Safety (KOFONS) grant funded by the Korean Government (Nuclear Safety and Security Commission, NSSC) [grant number RS-2021-KN066120] and a Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korean government (MOTIE) [grant number 20222020800130, Development and demonstration of hybrid power system using ORC (Organic Rankine Cycle) and TEG (Thermoelectric Generator) for low and medium temperature industrial waste heat recovery] and R&D Program of "A study on the improvement of cooling capacity and the coolant system for the DEMO divertor (Study of Demonstration Plant Design Concept and Base Technology, CN2301-11)" through the Korea Institute of Fusion Energy (KFE) funded by the Government funds.

Appendix A. Supplementary data

The following is the Supplementary data to this article:

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Multimedia component 1.

Recommended articles

Data availability

Data will be made available on request.

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